

NEW PRODUCTS

TECHNOLOGY

COMPONENTS & CHIPS

4-in-1 MEMS air quality sensor

Bosch Sensortec unveils the BME690, 4-in-1 MEMS gas sensor that measures gases, humidity, temperature, and barometric pressure. It reduces power

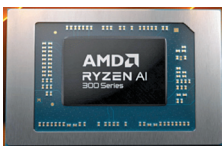
consumption by up to 50% compared to its predecessor, making it ideal for battery-operated devices. The sensor aligns with WELL and RESET standards for indoor air quality. Maintaining pin compatibility with previous models, it offers precise monitoring, AI capabilities, and is perfect for indoor air quality, breath analysis, and food freshness detection.

BOSCH

<https://www.bosch-sensortec.com>

Advanced AI processors

The AMD Ryzen AI 300 series processors, with the world's most powerful neural processing unit (NPU), will transform laptop computing. The AMD Ryzen 9000 series processors reinforce



AMD's leadership in performance and efficiency for desktops. Built on XDNA 2, the AI 300 series delivers 50 TOPS of AI power with up to 12 'Zen 5' CPU cores and 50% more L3 cache. The Ryzen 9000 series, also based on 'Zen 5,' offers 16% better IPC performance.

AMD

<https://www.amd.com/en.html>

Energy-efficient MCUs

Nuvoton Technology unveiled the ARM Cortex-M23 M2L31 microcontroller (MCU) series, designed for energy-efficient, high-performance



embedded computing. Running up to 72MHz, it offers 64 to 512k bytes of ReRAM and 40 to 168k bytes of SRAM. Features include two CAN FD and two USB Type-C PD 3.0 ports, operating voltage of 1.71V-3.6V, and temperature range of -40°C to 105°C. With a normal operating current of 60µA/MHz and 0.5µA in deep shutdown mode, it is ideal for battery management, industrial automation, and consumer devices.

Nuvoton Technology

<https://www.nuvoton.com>

AI-connected MCUs

Infineon Technologies launched the AI-ROC CYW5591x connected microcontroller (MCU) family, integrating Wi-Fi 6/6E and Bluetooth low energy 5.4,



for cost-efficient, power-saving smart home, industrial, wearable, and IoT devices. It features ModusToolbox software, RTOS, Linux drivers, validated Bluetooth stack, sample code, Matter software, and partner network support. The CYW55913, CYW55912, and CYW55911 models offer various band options, up to +24dBm Wi-Fi transmit power, matter-over-Wi-Fi, +19dBm

Bluetooth transmit power, ARM Cortex M33 MCU, and robust security features.

Infineon Technologies

<https://www.infineon.com>

64-bit MPUs

Microchip Technology's new PIC64 family of 64-bit MPUs meets the growing demands of real-time, compute-intensive applications like smart embedded vision and machine learning. Designed for power efficiency,



hardware-level security, and high reliability, these MPUs cater to sectors like industrial, automotive, IoT, aerospace, and defence. Featuring a 64-bit RISC-V quad-core processor with asymmetric multiprocessing, they support Linux, real-time OS, and bare metal applications. The PIC64-HPSC family enhances space computing, developed with NASA-JPL, ensuring radiation and fault tolerance for aerospace and defence applications.

Microchip Technology

<https://www.microchip.com>

LTE Cat 1bis cellular module

u-blox expands its R10 product range to meet rising LTE Cat 1bis cellular connectivity demand, projected to dominate 43.6% of non-handset cellular devices by 2029. Key features



include the ultra-compact 16x16 mm LEXI-R10 Global, ideal for IoT applications like trackers and wearables, and the SARA-R10 series. Both